

KOKPEKTY, Kazakhstan

WMO: 365350

Lat: 48.750N

Lon: 82.367E

Elev: 1680

StdP: 13.83

Time Zone: 6.00 (E06)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-34.0	-29.2	-38.5	0.6	-33.6	-33.9	0.9	-28.9	18.9	2.3	17.3	3.3	1.1	310	0.574

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	23.7	90.3	64.6	87.1	63.6	83.9	62.5	67.2	83.9	65.6	81.9	64.1	80.3	6.3	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
61.4	86.4	74.4	59.2	79.8	73.4	57.2	74.4	72.2	32.7	84.0	31.4	81.8	30.2	80.4	78.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 19.4	16.8	13.8	DB	-38.8	95.5	5.9	3.0	-43.1	97.7	-46.5	99.4	-49.8	101.1	-54.1	103.3	(4)
(5)			WB	-38.9	71.2	5.9	2.7	-43.1	73.2	-46.5	74.7	-49.8	76.2	-54.1	78.2	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	36.1	-5.0	1.4	18.0	42.2	55.6	66.2	69.3	66.4	54.1	39.6	20.9	3.0		
	DBStd	28.00	13.17	12.04	12.73	9.88	7.38	6.39	5.40	6.09	7.99	8.14	12.96	12.49		
	HDD50	7064	1704	1362	993	263	33	1	0	0	49	330	872	1457		
	HDD65	10896	2169	1782	1458	685	299	62	20	56	335	786	1322	1922		
	CDD50	2006	0	0	0	28	206	487	598	508	171	8	0	0		
	CDD65	365	0	0	0	0	8	98	153	99	7	0	0	0		
	CDH74	4882	0	0	0	7	240	1262	1679	1449	245	0	0	0		
	CDH80	1664	0	0	0	0	48	414	606	538	58	0	0	0		
	WSAvg	4.2	2.6	2.9	3.6	5.4	5.8	5.1	4.5	4.7	4.8	4.4	3.7	2.8		
	PrecAvg	12.4	1.1	0.8	0.6	0.7	1.3	1.2	1.4	0.8	0.7	0.9	1.5	1.4		
(15) Precipitation	PrecMax	21.4	3.1	3.1	2.6	1.9	5.8	3.5	5.2	3.0	2.4	2.0	4.1	3.5		
	PrecMin	6.9	0.2	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.2		
	PrecStd	3.0	0.7	0.6	0.5	0.5	1.1	0.8	1.1	0.7	0.6	0.6	1.0	0.7		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	26.0	30.6	48.4	74.0	85.6	91.5	94.6	94.6	86.6	70.4	51.2	32.4		
		MCWB	24.4	27.9	41.5	56.9	63.2	64.6	66.3	66.2	62.6	53.8	41.9	29.3		
	2%	DB	22.6	26.3	42.2	68.9	79.8	88.1	90.2	89.9	80.2	64.4	45.1	27.7		
		MCWB	21.1	23.7	37.0	54.2	59.8	63.6	64.8	64.5	60.3	51.5	38.3	25.6		
	5%	DB	19.2	22.9	38.3	64.5	76.1	85.0	86.5	86.2	76.2	59.0	40.3	24.0		
		MCWB	17.7	20.8	33.9	51.6	58.2	62.4	63.8	63.4	58.8	48.3	35.4	21.9		
	10%	DB	15.6	20.0	35.2	59.6	72.1	81.5	83.0	82.1	72.0	54.6	36.3	21.1		
		MCWB	14.2	18.0	31.5	48.6	56.6	61.3	62.6	62.1	56.8	45.6	32.4	19.4		
	0.4%	WB	25.0	27.9	41.4	58.4	65.0	68.8	69.7	69.3	64.7	55.0	42.6	29.6		
		MCDB	25.9	30.5	47.5	72.5	81.0	84.2	86.1	88.7	82.8	67.4	49.8	32.2		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	21.3	23.8	37.4	54.8	61.4	65.9	67.5	66.5	61.6	51.7	38.7	25.7		
		MCDB	22.5	26.0	41.9	67.7	76.3	82.4	82.9	84.9	78.0	63.4	44.2	27.6		
	5%	WB	17.8	20.8	34.2	52.0	59.2	64.2	65.8	64.6	59.1	48.8	35.6	22.0		
		MCDB	19.3	22.8	38.0	63.2	73.8	80.6	80.9	82.8	75.2	58.2	40.2	23.8		
	10%	WB	14.3	18.0	31.7	49.1	57.2	62.5	64.3	62.9	56.9	46.1	32.6	19.4		
		MCDB	15.6	20.0	35.2	58.8	70.7	78.4	79.5	79.9	71.3	54.0	36.3	21.1		
(35) Mean Daily Temperature Range	5% DB	MDBR	21.0	22.8	20.7	21.9	25.3	25.3	23.7	26.8	28.4	22.9	16.8	18.7		
		MCDBR	16.8	20.2	18.3	31.8	31.8	30.2	29.1	33.7	34.7	32.1	20.8	15.6		
		MCWBR	15.9	18.5	14.5	19.7	16.8	12.9	11.8	14.8	19.3	21.5	15.7	14.3		
		MCDBR	16.5	19.5	17.4	29.3	28.8	27.7	25.4	29.8	33.4	29.9	20.1	15.3		
	5% WB	MCWBR	15.6	18.0	14.0	19.0	16.8	14.1	12.9	15.2	19.8	20.8	15.4	13.9		
(40) Clear-Sky Solar Irradiance	taub		0.289	0.320	0.370	0.410	0.397	0.402	0.419	0.384	0.345	0.346	0.323	0.293		
	taud		2.080	2.023	2.026	2.120	2.219	2.277	2.242	2.318	2.374	2.280	2.161	2.063		
	Ebn at Noon		234	254	261	265	274	273	266	271	271	246	220	212		
	Edh at Noon		28	38	45	45	43	41	42	37	31	29	26	25		
(44) All-Sky Solar Radiation	RadAvg		534	899	1316	1617	2046	2162	2014	1855	1425	824	464	395		
	RadStd		29	31	97	118	116	132	136	116	104	78	48	34		

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	-1.37	-1.49	N/A	N/A	N/A	N/A	
(47)	Regional (5 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page